

Sana'a University

Faculty of Computer and Information Technology

Department of Cyber Security

RESEARCH PROPOSAL

A Hybrid Methodology for Integrating Real-Time Attack Surface Management with Network Traffic Analysis to Generate AI-Enhanced Detection Rules for Reducing Attack Detection Time in Security Operations Centers

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Abstract

Security Operations Centres (SOCs) are central to the monitoring, detection, analysis and response to cyber threats. Despite their importance, modern SOCs continue to grapple with a number of issues including delayed detection of attacks, high false positive rates, alert fatigue and a lack of correlation between asset exposure information and actual network behaviour. Current detection methods are often reliant on static signatures, generic correlation rules or isolated alerts which may not provide sufficient context to detect multi-stage or evolving attacks. This research proposes a hybrid methodology which combines real time Attack Surface Management (ASM) data with network traffic analysis to produce AI enhanced dynamic detection rules. The approach involves combining ASM data (asset exposure, open ports, running services, vulnerabilities, asset criticality) with network telemetry (source and destination IP addresses, ports, protocols, connection frequency, packet count, byte count and time window behaviour) to identify high risk cases where suspicious traffic is directed at exposed, vulnerable or critical assets. AI is used as an auxiliary component for risk scoring, prioritisation and rule generation. The intention is not to replace the SOC analyst or to replace existing SIEM and IDS/IPS platforms. It is to provide a structured and interpretable approach which can be used by analysts to produce more context rich detection rules. The methodology will be evaluated against existing static detection rules using a range of metrics including MTTD, false positive rate, precision, recall, F1 score, detection rate and rule interpretability. The contribution of the research is expected to be a SOC focused methodology which provides improved detection context, reduces detection time, enables reduction in false positives and produces dynamic detection rules which are based on both attack surface visibility and observed network behaviour.

Keywords

Security Operations Centre; Attack Surface Management; Network Traffic Analysis; SIEM; IDS/IPS; Artificial Intelligence; Dynamic Detection Rules; Mean Time to Detect; False Positive Reduction; Cybersecurity.

Executive Research Snapshot

Item	Description
Proposed Title	A Hybrid Methodology for integrating real time attack surface mapping with network traffic analysis to generate AI enhanced detection rules to reduce attack detection times in SOC centres.
Research Area	Cybersecurity, Security Operations Centres, Attack Surface Management, Network Traffic Analysis, Intrusion Detection, AI enhanced detection engineering.
Research Type	Quantitative Computing Research.
Study Family	Algorithm/System Evaluation.
Core Problem	Current SOC detection methodologies are often reliant on static rules, isolated alerts and a lack of correlation between asset exposure and real time network activity which can result in increased detection times and false positives.
Main Research Gap	There is a lack of a hybrid SOC focused methodology that combines real time ASM data with network traffic analysis and uses AI to produce interpretable and dynamic detection rules.
Proposed Solution	A hybrid methodology that brings together ASM data, network traffic features, correlation logic and AI enhanced risk scoring to produce contextual detection rules for the SOC analyst.
Input Data	ASM data (assets, ports, services, exposure status, vulnerabilities, asset criticality) and network traffic data (source IP, destination IP, ports, protocols, connection counts, packet counts, byte counts and time windows).
Main Techniques	Data pre-processing, feature extraction, ASM to traffic correlation, AI enhanced risk scoring, dynamic rule generation, baseline comparison and quantitative evaluation.

Expected Evaluation Protocol	Comparison of traditional static detection rules against the proposed hybrid detection methodology using controlled datasets, simulated scenarios and lab network data.
Metrics to be evaluated	Mean Time to Detect, Detection Rate, False Positive Rate, Precision, Recall, F1-score and Rule Interpretability.
Expected contribution	A structured SOC oriented methodology that improves detection context, reduces detection time, enables reduction in false positives and results in more interpretable AI enhanced detection rules.
Practical value	The methodology would assist SOC analysts by enabling them to focus on suspicious activity that is targeting exposed, vulnerable or critical assets and would also assist in the creation of more relevant detection rules.
Scope	The study is focused on the generation of defensive detection rules using authorised ASM and network traffic data only. It does not cover offensive exploitation, unauthorized scanning, payload decryption or full production SOC deployment.

Alignment with Supervisor's Chapter Guides

This is a quantitative computing research design. The topic falls within the algorithm/system evaluation family as it's concerned with the design and evaluation of a hybrid SOC oriented methodology using ASM data, network traffic logs, AI enhanced risk scoring, dynamic detection rule generation and a range of evaluable metrics.

Supervisor's Guide Requirement	**Alignment in This Proposal**
Problem, Purpose, and Research Questions	Addressed in Chapter 1 via the research background, the rationale for the study, the research problem, the aim, the objectives, the research questions and the hypotheses.
Literature review and research gap.	Chapter 2 looks at the topic of SOC detection via a focused review of SOC detection, ASM, network traffic analysis, IDS/IPS and SIEM rules, AI based detection and dynamic rule generation.
Quantitative Research Design	Chapter 3 deals with the quantitative research design through the use of an algorithm/system evaluation design based on measurable detection performance.
Data Source Selection	The data source selection is covered in Chapter 3 with the use of ASM data and network traffic data as the primary input sources.
Unit of Analysis and Observation	The unit of analysis and observation is dealt with in Chapter 3 via the use of correlated security scenarios which combine asset exposure with observed network traffic behaviour.
Experimental Environment	The experimental environment is described in Chapter 3 with details of the tools, datasets, preprocessing, programming environment and evaluation scripts.

Experimental Protocol	The experimental protocol is also covered in Chapter 3 with baseline comparison, data pre-processing, ASM to traffic correlation, AI enhanced risk scoring and rule generation.
Evaluation Metrics	Evaluation metrics covered in Chapter 3 are Mean Time to Detect, detection rate, false positive rate, precision, recall, F1-score and rule interpretability.
Validity and Reproducibility	Validity and reproducibility in Chapter 3 are covered via threats to validity, assumptions, dataset documentation and experimental environment documentation.
Ethical Considerations	Ethical considerations in Chapter 3 are that no scanning of networks without authorisation, no exploitation and no collection of private network data is carried out.
Expected Contribution	The expected contribution is addressed in Chapters 1 and 3 via the proposed hybrid methodology for the generation of contextual and dynamic SOC detection rules.

List of Abbreviations

Abbreviation	Meaning
AI	Artificial Intelligence
ASM	Attack Surface Management
ATT&CK	Adversarial Tactics, Techniques and Common Knowledge
C2	Command and Control
CSV	Comma Separated Values
F1-score	Harmonic Mean of Precision and Recall
FPR	False Positive Rate
IDS	Intrusion Detection System
IP	Internet Protocol
IPS	Intrusion Prevention System
ML	Machine Learning
MTTD	Mean Time to Detect
NTA	Network Traffic Analysis
PCAP	Packet Capture
RDP	Remote Desktop Protocol
SIEM	Security Information and Event Management
SOC	Security Operations Centre
SSH	Secure Shell
TCP	Transmission Control Protocol
UDP	User Datagram Protocol
VPN	Virtual Private Network

Contents

Abstract i

Keywords ii

Executive Research Snapshot iii

Alignment with Supervisor's Chapter Guides v

List of Abbreviations vii

List of Tables xi

1 Introduction and Research Foundation 1

1.1 Background and Context 1
1.2 Research Motivation 1
1.3 Research Problem 2
1.4 Problem Decomposition 2
1.5 Research Gap 2
1.6 Research Aim 4
1.7 Research Objectives 4
1.8 Research Questions 4
1.10 Expected Contribution and Novelty 5
1.12 Scope and Delimitations 6
1.13 Limitations 6
1.14 Operational Definition of Key Terms 6
1.15 Organization of the Proposal 6

2 Literature Review and Research Gap Development 9

2.1 Introduction 9
2.2 Security Operations Centers and Detection Challenges 9
2.3 Attack Surface Management 10
2.4 Network Traffic Analysis and Network Telemetry 10

2.5 IDS/IPS and SIEM Detection Rules . . . 11
2.6 Artificial Intelligence in Cybersecurity Detection . . . 11
2.7 Dynamic Detection Rule Generation . . . 11
2.8 Summary of Related Research Areas . . . 12
2.9 Research Gap Development . . . 12
2.10 Conceptual Framework of the Proposed Study . . . 12
2.11 Chapter Summary . . . 14

3 Research Methodology and Experimental Design 15
3.1 Introduction . . . 15
3.2 Research Design . . . 15
3.3 Proposed Framework Overview . . . 16
3.4 Data Sources . . . 16
3.4.1 Attack Surface Management Data . . . 17
3.4.2 Network Traffic Data . . . 17
3.5 Data Preprocessing . . . 18
3.6 Feature Selection . . . 18
3.7 Correlation Logic . . . 18
3.8 AI-Enhanced Risk Scoring . . . 19
3.9 Dynamic Detection Rule Generation . . . 20
3.10 Baseline Comparison . . . 20
3.11 Experimental Protocol . . . 20
3.12 Evaluation Metrics . . . 22
3.13 Experimental Environment . . . 22
3.14 Ethical Considerations . . . 23
3.15 Threats to Validity . . . 23
3.16 Experimental Variables . . . 23
3.17 Proposed Timeline . . . 23
3.18 Expected Outcomes . . . 25
3.19 Chapter Summary . . . 25

Conclusion 27

Bibliography 28

A ASM Data Template 30

B Network Traffic Data Template 31

C Dynamic Detection Rule Template 32

D Evaluation Checklist **34**

E Risk Register Template **35**

List of Tables

1.1 Problem Decomposition Summary 3
1.2 Scope and Delimitations 6
1.3 Operational Definition of Key Terms 7

2.1 Summary of Related Research Areas 13
2.2 Research Gap Development Matrix 13

3.1 Feature Categories Used in the Proposed Methodology 19
3.2 Baseline Comparison Plan 20
3.3 Evaluation Metrics 22
3.4 Experimental Variables 24
3.5 Proposed Research Timeline 25

Chapter 1

Introduction and Research Foundation

1.1 Background and Context

Organisations today operate in a highly interconnected digital world with networks, cloud services, remote access, web apps, endpoints and a whole host of third party integrations all contributing to a vastly expanded attack surface. The result is that the threats facing organisations are increasingly sophisticated, adaptive and difficult to detect using traditional monitoring techniques alone. Modern cyber risk management requires an organisation to have a clear understanding of its assets, its risks, its vulnerabilities, its threats and its security outcomes [1]. The Security Operations Centre (SOC) is at the heart of the organisation's security posture and is responsible for the continuous monitoring of security events, the analysis of suspicious activity, the detection of threats and the provision of support for incident response. Continuous monitoring is a necessity as an organisation needs to have an ongoing awareness of its information security, its vulnerabilities, its threats and the effectiveness of its controls [2]. The SOC will typically be leveraging a range of security technologies to achieve this including Intrusion Detection and Prevention Systems (IDS/IPS), Security Information and Event Management (SIEM) systems, firewalls, endpoint detection systems, vulnerability scanners and threat intelligence. IDS/IPS are a key element of network security monitoring and are used to detect and prevent potentially malicious or suspicious activity across a range of deployment scenarios [3]. Effective log management is also a key enabler of security monitoring, incident investigation and event correlation within the SOC environment [4].

1.2 Research Motivation

The raison d'etre for this work is the growing chasm between the sophistication of modern cyber attacks and the ability of the traditional SOC detection methodologies to detect them rapidly and effectively. Modern attackers typically employ multi-stage attacks which start off with reconnaissance and exploitation of exposed assets, followed by credential abuse and lateral

movement.

Ment, command and control comms or data exfiltration. Reducing the time to detection is also important from an incident response standpoint as effective incident response is predicated on timely detection, analysis, containment, eradication and recovery [6].

1.3 Research Problem

Despite the huge strides that have been made in the area of cyber security monitoring, many SOC's are still unable to detect modern and sophisticated attacks in a timely and accurate fashion. The current state of the art in detection is largely based on the use of signatures, static SIEM correlation rules or isolated alerts from individual tools. The research problem is therefore that there is no integrated, time-aware approach to detection that combines Attack Surface Management with network traffic analysis and uses AI to generate interpretable, dynamic detection rules which would reduce detection time and improve the quality of the alerts. ML and anomaly based detection may be of some assistance but previous work has shown that IDS's continue to suffer from the problems of false positives, operational deployment and the unrealistic closed world assumption [9].

1.4 Problem Decomposition

To clarify the research problem we break it down into a number of constituent parts. Each of these parts represents a limitation in the current state of the art in SOC detection and each has a direct bearing on the proposed hybrid approach.

1.5 Research Gap

There's been a lot of progress in the areas of intrusion detection, network traffic analysis, SIEM correlation, vulnerability management and AI in cybersecurity. The research gap that this paper addresses is that these are all typically treated as separate functions. The paper proposes a hybrid SOC oriented approach that brings together real time Attack Surface Management with network traffic analysis and uses AI to produce dynamic detection rules that are both interpretable and useful in an operational context. The guidance on risk assessment is that you need to have a clear understanding of the threats, the vulnerabilities, the likelihood and the impact before you can make any cybersecurity decisions [10]. That underpins the need to bring together the asset exposure and vulnerability context with the actual network behaviour that you are observing.

Table 1.1: Problem Decomposition Summary

Problem Component	**Description**	**Research Implication**
Attack Surface Visibility Gap	SOC detection systems may not be making full use of the information available on exposed assets, open services, vulnerable systems and attack paths.	The proposed methodology must have ASM data as a core input to the rule generation process.
Network Behavior Isolation	Network traffic analysis can pick up on anomalous patterns without any understanding of the exposure or criticality of the destination asset.	The methodology has to link traffic behaviour with asset exposure and risk context.
Static detection rules	(many of the IDS/SIEM rules are pre-defined) will not be able to adapt quickly to changes in your assets, services or threats.	The study must be about dynamic rule generation based on the current attack surface and traffic evidence.
Slow detection time	is a problem due to the weak correlation between ASM and network telemetry which can result in a delay in the identification of actual attacks.	The methodology has to be geared towards reducing Mean Time to Detect via contextual and early detection logic.
High False Positive Rate	Alerts generated without any context at all can easily overwhelm the SOC analysts.	The methodology must improve alert accuracy via exposure aware and behaviour aware rule conditions.
Limited AI utilisation:	AI can be used for classification but not necessarily for the generation of explainable rules or for the creation of detection logic that is SOC oriented.	The study must specify how AI is used to support risk scoring, risk prioritisation and the generation of interpretable detection rules.

1.6 Research Aim

This study is all about designing and evaluating a hybrid approach that combines Attack Surface Management with network traffic analysis in order to produce AI enhanced dynamic detection rules which will in turn reduce the time to detect an attack and improve the accuracy of the alerts in the SOC.

1.7 Research Objectives

To achieve the research aim, this study will pursue the following objectives:

1. To look at the limitations of the current range of SOC detection techniques with a particular focus on static rules, the delay in detection, the issue of false positives and the poor integration between ASM and network traffic analysis.
2. To work out which types of ASM data are useful for contextual detection (exposed assets, open ports, running services, vulnerabilities, asset criticality, exposure status etc).
3. To find the right network traffic attributes to use as indicators of suspicious behaviour (i.e. connection frequency, source/destination, port activity, protocol usage, flow duration, packet and byte counts, time window behaviour etc).
4. To design a correlation mechanism that correlates ASM information with network traffic behaviour in order to identify high risk attack scenarios.
5. To develop an AI enhanced risk scoring component to support alert prioritisation and dynamic detection rule generation.
6. To produce interpretable detection rules based on correlated ASM and traffic evidence.
7. To benchmark the proposed methodology against traditional static detection rules or against baseline detection logic.
8. To assess the proposed methodology using the usual metrics of Mean Time to Detect, false positive rate, precision, recall, F1-score and rule interpretability.

1.8 Research Questions

This study is guided by the following research questions:

1. What are the primary limitations of the current SOC detection approaches in terms of correlating attack surface information with real time network traffic behaviour?

2. Which ASM attributes are most useful for supporting contextual and dynamic deteWhich ASM attributes are most useful for supporting contextual and dynamic detection rule generation?or related to exposed or vulnerable assets?

3. How can artificial intelligence support risk scoring and the generation of inteHow can AI be used to support risk scoring and the creation of interpretable detection rules?me compared with traditional static detection rules?

4. To what extent can the proposed methodology reduce false positives and improve To what extent can the proposed methodology reduce the number of false positives and improve the accuracy of the alerts in a SOC?luation design, the following hypotheses are proposed:

5. H1: Integrating ASM data with network traffic analysis improves attack detectioH1: Combining ASM data with network traffic analysis provides better attack detection context than network traffic analysis alone.ties target- ing exposed or vulnerable assets.

6. H3: Dynamic detection rules generated from correlated ASM and traffic evidence H3: Dynamic detection rules created from correlated ASM and traffic data result in a lower Mean Time to Detect than would be achieved with conventional static detection rules.generic detection rules that do not use attack surface context.

1.10 Expected Contribution and Novelty

The expected contribution of this study is a structured hybrid methodology for The expected contribution of the study is a structured hybrid approach to SOC detection via the integration of real time Attack Surface Management, network traffic analysis and AI enhanced rule generation. The novelty of the study is in the way it brings together three elements that are typically considered in isolation; ASM, network telemetry and detection rule generation. Rather than using ASM purely for vulnerability management and network telemetry purely for anomaly detection, the proposed methodology uses both to generate context aware and dynamic rules.ed from scientific, practical, and operational perspectives. Scientifically, the study contributes to cybersecurity research by proposing a hybrid methodology that connects attack surface visibility with behavioral network analysis and AIsupported rule generation. Practically, the proposed methodology may help organizations improve detection quality by generating rules that are aligned with their actual attack surface and observed traffic behavior. Operationally, the study may support SOC analysts by reducing alert overload, improving prioritization, and shortening the time needed to identify high-risk activities.

1.12 Scope and Delimitations

The scope of this work is the design and evaluation of a hybrid approach to the generation of AI-enhanced detection rules using ASM data and network traffic analysis.

Table 1.2: Scope and Delimitations

Included in the study,	**excluded from the study:**
Attack Surface Management data (assets, ports, services, exposure, vulnerabilities).	Full enterprise scale ASM platform development.
Network traffic logs or flow level records.	Payload decryption or deep packet inspection of the encrypted content.
AI based risk scoring and rule generation support.	Fully autonomous rule deployment with no analyst review required.
Dynamic and interpretable detection rule generation.	Offensive exploitation or unauthorised attack execution.
Evaluation using measurable SOC metrics.	Replacement of existing SIEM, IDS or SOC platforms.
Controlled experimental (or lab based) evaluation.	Full production SOC deployment.

1.13 Limitations

Several limitations are to be expected. Firstly the work may be based on lab generated data, public datasets or controlled network traffic. That won't necessarily be a true reflection of the complexity and noise of a real enterprise SOC. Secondly the quality of the ASM data will impact the efficacy of the approach. Poor asset inventory, out of date vulnerability data or incomplete exposure data will all have an impact on the quality of the rules that are generated.

Thirdly AI based scoring will probably need to be carefully calibrated to avoid either over or under estimating the risk. Poorly calibrated risk scores could result in ineffective prioritisation or an excessive number of alerts.

1.14 Operational Definition of Key Terms

1.15 Organization of the Proposal

The proposal is in three main parts. Part 1 is the introduction and research base (background, motivation, research problem, problem breakdown, research gap, aim, objectives, research questions, hypotheses, expected contribution, significance, scope, limitations and operational definitions). Part 2 is the literature review and research gap development. This covers SOC

Table 1.3: Operational Definition of Key Terms

Term	**Operational Definition in this study:**
SOC (Security Operations Centre)	is a security operations environment responsible for the monitoring, detection, analysis and response to cyber threats.
Attack Surface Management	The process of identifying and monitoring your assets, services, ports, vulnerabilities and exposures that could be the target of an attack.
Real-time attack surface mapping.	A continuously updated view of the assets, services, vulnerabilities and risk states that are exposed.
Network Traffic Analysis:	the analysis of communication behaviour using logs, flows, packets or metadata (IP addresses, ports, protocols, counts, timings etc).
Network Telemetry:	security relevant data collected from network devices, flows, firewalls, IDS/IPS and/or monitoring sensors.
Dynamic Detection Rule:	a detection rule which is created or modified as a result of the current ASM data and observed network behaviour.
AI-Enhanced Detection:	the use of AI or machine learning to assist with risk scoring, classification, prioritisation or rule generation.
False Positive:	an alert which has identified benign activity as malicious or suspicious.
Mean Time to Detect (MTTD):	the average time it takes to detect an attack or suspicious activity once it has commenced.
Rule Interpretability:	the extent to which a generated detection rule can be understood, reviewed and validated by a SOC analyst.

operations, Attack Surface Management, network traffic analysis, IDS/IPS and SIEM detection rules, AI based threat detection, dynamic rule generation, alert prioritisation, false positive reduction and Mean Time to Detect.

Chapter 3 is all about the research methodology and experimental design. It details the proposed framework architecture, the ASM data collection, the network traffic data collection and preprocessing, the correlation logic, the AI based risk scoring, the dynamic detection rule generation, the baseline comparison, the evaluation metrics, the experimental environment, the ethical considerations, the threats to validity, the expected outcomes and the proposed timeline.

Chapter 2

Literature Review and Research Gap Development

2.1 Introduction

This chapter provides a review of the key areas of research relevant to the study. The purpose of the review is to establish the theoretical and practical basis for a hybrid approach that will combine Attack Surface Management (ASM), network traffic analysis and artificial intelligence in order to produce dynamic detection rules for use in a Security Operations Centre (SOC) environment. The review is structured around six key themes: SOC detection challenges, Attack Surface Management, network traffic analysis, IDS/IPS and SIEM rule based detection, AI enhanced detection and dynamic detection rule generation. The review of these themes serves to identify the research gap that the proposed study seeks to address.

2.2 Security Operations Centers and Detection Challenges

SOCS are responsible for monitoring, detecting, analysing and responding to cyber threats. Modern SOCs gather security relevant events from a wide variety of sources including firewalls, IDS/IPS, endpoint agents, vulnerability scanners, SIEMs and threat intelligence feeds. Effective security monitoring requires a continuous view of the assets, threats, vulnerabilities and the effectiveness of the security controls within the organisation [1]. Log management is also a fundamental function within a SOC. Security logs are used for detection, investigation, analysis and response. There are of course challenges with logs too in that they can be in huge volumes, in inconsistent formats, difficult to store, difficult to review and difficult to correlate [2]. The sheer volume of logs and the difficulties associated with their management can make it very difficult for the SOC analysts to detect an attack in a timely and accurate manner. The tools available to the SOC do provide a significant amount of visibility but the SOC environment is still going to face a number of operational challenges.

The operational challenges are: analysts will be deluged with a large number of alerts, many of which will be false positives. The detection rules will be static and may not reflect the current state of exposure of the organisation's assets. The security tools will be operating in isolation and so will not provide a complete picture of the attack.

2.3 Attack Surface Management

Attack Surface Management is the process of identifying and monitoring assets, services, ports, vulnerabilities and exposures that an attacker could potentially target. In the context of cyber risk management, an organisation will want to understand, assess, prioritise and communicate its cyber risk [3]. ASM is used to provide that visibility into the exposed systems and potential attack vectors. External Attack Surface Management is also identified in practical guidance as a means of identifying and managing the internet facing assets and exposures that could be targeted by an attacker [4]. The ASM data in this study could include asset IP addresses, asset names, exposed ports, running services, service versions, vulnerability severity, exposure status and asset criticality. This information is useful in the context of detection as activity that is suspicious in the sense of targeting an exposed or vulnerable asset is likely to be of higher risk than the same activity targeting a non-critical or non-exposed asset. ASM is however often used primarily in the context of vulnerability management or asset visibility and is less commonly used in a way where it is directly integrated with real time network traffic analysis to enable dynamic detection rule generation. There is therefore an opportunity to use ASM as a contextual input to detection in the SOC.

2.4 Network Traffic Analysis and Network Telemetry

Network traffic analysis is all about observing the behaviour of traffic between systems. You can do this via packet captures, flow records, firewall logs, IDS alerts, NetFlow-like data or SIEM events. Useful traffic attributes might be source IP, destination IP, destination port, protocol, connection count, byte count, packet count, flow duration, time window and connection frequency. Network telemetry is important in that it tells you what's actually happening in the network. For instance, repeated connection attempts to SSH, RDP, VPN or web services could be indicative of reconnaissance, brute force, exploitation or some form of anomalous access. The problem with traffic analysis alone is that it may not provide sufficient context. A given traffic pattern may be suspicious but its actual risk is going to be determined by the exposure, the vulnerability and the criticality of the asset at the destination. So traffic behaviour in conjunction with ASM data can be a useful enabler of...

more accurate and context aware detection.

2.5 IDS/IPS and SIEM Detection Rules

IDS/IPS and SIEM platforms will typically use rules to detect suspicious or malicious activity. These rules can be based on signatures, thresholds, event correlation, indicators of compromise or known attack patterns. The advantage of rule based detection is that the rules are transparent and can be reviewed by the SOC analysts. MITRE ATT&CK provides a structured knowledge base of adversary tactics and techniques which can be used to support threat informed detection engineering and the mapping of attack behaviour [5]. This kind of knowledge base can assist the SOC in understanding attacker behaviour and in designing detection that is aligned to the techniques of the adversary. The downside of traditional rules is that they are often hand crafted, static and generic. A generic rule may not take into account whether the asset in question is exposed, vulnerable or critical. This can lead to an increase in false positives or a delay in the detection of high risk activity. The aim of the study is to address this by using ASM and network traffic analysis in combination to produce more contextual and dynamic detection rules.

2.6 Artificial Intelligence in Cybersecurity Detection

AI and machine learning have been extensively researched in the field of cybersecurity for a number of applications including classification, anomaly detection, clustering, risk scoring and alert prioritisation. Buczak and Guven [6] have reviewed the use of data mining and machine learning for cyber security intrusion detection and their work clearly demonstrates the value of a data driven approach to cyber analytics. Anomaly detection is also a relevant area of study for SOC's as attacks can manifest as anomalies in behaviour. Chandola, Banerjee and Kumar [7] have provided a very broad review of anomaly detection techniques and this underpins the use of behavioural analysis in the detection of suspicious activity. The application of machine learning to network intrusion detection is however not without its problems. Sommer and Paxson [8] point out a number of issues with the application of machine learning to intrusion detection, including the use of unrealistic assumptions and the difficulty of operational deployment. For these reasons this study does not see AI as a replacement for SOC analysts. AI is used as a tool to support the risk scoring process and to assist in the generation of interpretable detection rules.

2.7 Dynamic Detection Rule Generation

Dynamic detection rule generation is the creation or updating of detection rules based on the current security context rather than just using a set of static pre-defined rules. In the study in question the rules are generated on the basis of the relationship between the ASM data and the observed network traffic. The process can be summarised as:

1. Collect ASM data on assets, services, ports, exposure and vulnerabilities.
2. Collect network traffic (or network telemetry).
3. Correlate traffic behaviour with ASM context.
4. Calculate a risk score using AI or rule based logic.
5. Generate an interpretable detection rule.
6. Evaluate the generated rule using SOC relevant metrics.

This could lead to better detection as the rules are derived from the actual attack surface and the actual traffic profile of the organisation.

2.8 Summary of Related Research Areas

Table 2.1 sets out the principal research areas associated with this study.

2.9 Research Gap Development

The areas under review demonstrate that SOCs are already deploying a range of significant security technologies (SIEMs, IDS/IPS, ASM, network monitoring etc). The problem is that they are typically deployed in isolation. ASM gives you information about the assets that are exposed and their vulnerabilities. Network traffic analysis gives you information about the actual behaviour of the traffic. If you don't integrate those two sources of information then your detection system is going to be missing the context. The key research gap is in the absence of a hybrid SOC-centric methodology that takes real time ASM data and network traffic analysis and uses AI to generate interpretable and dynamic detection rules. That's a significant gap because detection shouldn't be reliant solely on generic signatures or static thresholds. The detection rules should take into account both the risk of the asset in question and the actual behaviour of the traffic. Table 2.2 contains the research gap development matrix.

2.10 Conceptual Framework of the Proposed Study

The conceptual framework of the study is built upon three inputs and one output. The first input is the ASM data (which covers exposed assets, ports, services, vulnerabilities and asset criticality). The second input is network traffic data (which covers the communication behaviour). The third input is AI enhanced analysis (which is used to support risk scoring and prioritisation). The output is a set of interpretable detection rules which can be reviewed and consumed by the SOC analysts. That's the relationship in a nutshell.

Table 2.1: Summary of Related Research Areas

Research Area	**Main Focus**	**Relevance to This Study**
SOC Detection	Monitoring: alerts, logs, incidents, security events.	Provides the operational environment where the proposed methodology will be applied.
Attack Surface Management	Identifying exposed assets, open ports, services, vulnerabilities and risks.	Provides contextual information for prioritizing suspicious traffic and generating rules.
Network Traffic Analysis	Analyse communication behaviour using flows, logs, packets or metadata.	Provides real-time behavioral evidence about possible attacks.
IDS/IPS and SIEM Rules	Detecting suspicious activity via signatures, thresholds and correlation rules.	Delivers the rule-based detection output that is the target of the proposed approach.
AI	to support classification, anomaly detection, risk scoring and prioritisation.	Supports dynamic and context aware rule generation.
Dynamic Rule Generation	Creating or refining detection rules in response to evolving data and risk context.	Represents the primary output of the proposed methodology.

Table 2.2: Research Gap Development Matrix

Current Limitation	**Impact on SOC Detection**	**Proposed Direction**
ASM and traffic data are generally analysed in isolation.	Limited attack context and poor alert prioritisation.	Correlate the ASM data with network traffic behaviour.
Detection rules are generally static.	Rules cannot adapt to changes in assets, services and threats.	Create dynamic rules based on current exposure and traffic.
Generic rules can result in false positives.	Analysists can get alert fatigue and have delayed response.	Use asset exposure and traffic context to enhance rule specificity.
AI is generally used for classification only.	Limited support for explainable SOC oriented rule generation.	Use AI for risk scoring and for support with the generation of interpretable rules.
There will be a delay in detection time.	Attacks can progress before they are detected.	Assess the methodology using the Mean Time to Detect and alert accuracy metrics.

ASM Data + Network Traffic Analysis + AI-Based Risk Scoring = Dynamic Detection Rules

This framework supports the core assumption of the study: detection rules are more useful when they are derived from both attack surface context and real time traffic behaviour.

2.11 Chapter Summary

This chapter covers the key concepts underpinning the study, namely SOC detection challenges, Attack Surface Management (ASM), network traffic analysis, IDS/IPS and SIEM rules, AI based detection and dynamic detection rule generation. It identifies the research gap which is the lack of integration between ASM and network telemetry in the generation of contextual detection rules. The next chapter details the research methodology and experimental design. It describes how the ASM and network traffic data will be collected, correlated, processed and used to generate AI enhanced detection rules. It also details the metrics that will be used to evaluate the reduction in detection time and the accuracy of the alerts.

Chapter 3

Research Methodology and Experimental Design

3.1 Introduction

This chapter details the research methodology and experimental design of the study. The aim of the study is to design and evaluate a hybrid methodology which will combine real time Attack Surface Management (ASM) data with network traffic analysis in order to produce AI enhanced dynamic detection rules for a SOC. The chapter covers the research design, the proposed framework, the data sources, the data pre-processing, the correlation logic, the AI enhanced risk scoring, the detection rule generation, the baseline, the evaluation metrics, the experimental environment, the ethical considerations, the threats to validity and the expected outcomes.

3.2 Research Design

This is a quantitative computing research study based on algorithm and system evaluation. It's not a study based on surveys or interviews. It's about designing a technical framework and then evaluating that framework on the basis of measurable metrics. This is in keeping with the way security monitoring and log analysis is conducted where you collect data, process it, correlate it and evaluate it in order to support detection and response activities [2]. The methodology proposed will be evaluated by comparing two detection approaches:

1. A traditional approach based on static or generic detection rules.
2. The proposed hybrid approach which combines ASM data, network traffic analysis and AI enhanced risk scoring to produce dynamic detection rules.

The comparison will be on whether the proposed approach can reduce attack detection time, improve alert accuracy and reduce the number of false positives.

3.3 Proposed Framework Overview

The proposed framework consists of five main stages:

1. ASM data collection.
2. Network traffic data collection.
3. Data preprocessing and feature extraction.
4. Correlation between ASM data and network traffic behaviour.
5. AI enhanced risk scoring and dynamic detection rule generation.

The general concept of the framework is to determine whether the suspicious network behaviour is targeting exposed, vulnerable or critical assets. If a high-risk relationship is found then an interpretable detection rule is produced which can be reviewed and used by the SOC analysts.



Figure 3.1: Proposed Hybrid SOC Detection Architecture

Figure 3.1 shows the main components of the proposed framework and the interaction between ASM data, network traffic analysis, risk scoring, detection rule generation and SOC analyst feedback.

3.4 Data Sources

The study will use two broad categories of data: ASM data and network traffic data.

3.4.1 Attack Surface Management Data

ASM data is data on assets and services that could be the target of an attack. In this study the ASM data could be data on assets and services that could be the target of an attack.

- Asset IP address.
- Asset name.
- Open ports.
- Running services.
- Protocol type.
- Exposure status.
- Vulnerability severity.
- Asset criticality.

This data can be gathered from asset inventories, vulnerability scanning tools, port scanning tools or from manually created CSV files for the purposes of experimentation. The ASM data will be used to provide the security context for the traffic analysis phase. Guidance on information security testing and assessment supports the use of technical assessment techniques such as vulnerability scanning and network discovery to identify weaknesses and to validate the security posture [9]. The ASM data for this study can therefore be derived from the output of authorised scanning exercises, asset inventories or from controlled experimental templates.

3.4.2 Network Traffic Data

Network traffic data is data which represents the network communication behaviour. It can be gathered from firewall logs, IDS alerts, NetFlow-like records, packet captures or SIEM events. The traffic data in this study could be any of the above.

- Source IP address.
- Destination IP address.
- Destination port.
- Protocol.
- Number of connections.
- Flow duration.

- Packet count.
- Byte count.
- Time window.
- Traffic direction.

The network traffic will be analysed to look for any suspicious patterns such as multiple connection attempts, access to unusual ports, high connection frequency or otherwise odd communication behaviour.

3.5 Data Preprocessing

Before analysis, both ASM data and network traffic data will be cleaned and prepared. The preprocessing stage includes:

1. Remove the incomplete and duplicate records.
2. Normalise IP address and port formats.
3. Convert the timestamps into a standard format.
4. Aggregating the traffic records by time window.
5. Mapping of destination IP addresses and ports to ASM assets.
6. Encoding of the categorical variables (service type and exposure level).

This stage ensures that the ASM data and the network traffic data can be correctly correlated.

3.6 Feature Selection

The proposed methodology will make use of features from both the ASM and the network traffic. Table 3.1 summarises the feature types used in the study.

3.7 Correlation Logic

The heart of the proposed methodology is the correlation of ASM data with network traffic. It's looking to see if there's suspicious traffic directed at an exposed, vulnerable or critical asset. A simple example would be if the ASM data shows that a server has RDP exposed on port 3389 and the traffic data shows a number of connection attempts to that port from outside the network, that would be a high risk scenario. The correlation logic is simply:

Table 3.1: Feature Categories Used in the Proposed Methodology

Feature Category	**Example Features**	**Purpose**
ASM Features	(Asset criticality, exposed port, service type, vulnerability severity, exposure status).	To provide some context on the risk level of the target asset.
Traffic Features:	Source IP, destination IP, destination port, protocol, connection count, packet count, byte count, time window.	To represent observed network behaviour.
Correlation features:	match between traffic destination and exposed asset, access to high risk service, multiple attempts to a vulnerable service.	To identify high risk relationships between exposure and behaviour.
Risk Features:	Combined risk score, anomaly level, priority level.	To support alert prioritisation and rule creation.

Exposed Asset + Suspicious Traffic Behaviour = High Risk Detection Scenario.

The output of the correlation stage will be enriched records that contain both the asset information and the traffic behaviour.

3.8 AI-Enhanced Risk Scoring

The AI enhanced risk scoring element will assign a risk score to each correlated event or scenario. The scoring may be based on a number of factors including asset criticality, exposure status, vulnerability severity, number of connection attempts, service type, traffic direction and time window behaviour. Machine learning has been extensively researched in the area of intrusion detection and cybersecurity analytics [6]; its deployment in the real world however must be very carefully considered as intrusion detection environments are complex, dynamic and subject to false positives and evolving attack behaviour [8].

The output of this stage may classify each scenario into one of three levels:

- Low risk.
- Medium risk.
- High risk.

The AI element could be a machine learning classifier or a rule-based scoring model. The role of the AI in this study is not to supplant the SOC analyst at all, but to assist with prioritisation and rule generation.

3.9 Dynamic Detection Rule Generation

After risk scoring the system produces interpretable detection rules for high risk scenarios. These rules can be in a generic rule format, SIEM correlation logic or IDS style rule format. For instance if the system detects a number of attempts against an exposed SSH service it might generate a rule such as:

> Alert if the same source IP is attempting to connect to an exposed SSH service more than 20 times in a 60 second period.

The rules generated should be something that SOC analysts can understand and review. That's a key requirement of the SOC environment in that you need rules that are explainable rather than just black box rules.

3.10 Baseline Comparison

To assess the proposed methodology we will compare it to a baseline. The baseline approach could be one that uses static detection rules with no ASM context. The baseline comparison is essential as traditional IDS/IPS and SIEM based detection approaches will often be rule or signature based. Our proposed methodology incorporates asset exposure and risk context into the detection process [10]. The comparison will allow us to determine whether the addition of ASM data to the network traffic analysis actually improves detection. Table 3.2 details the comparison plan.

Table 3.2: Baseline Comparison Plan

Approach	**Description**	**Expected Limitation**
Baseline detection	uses static or generic rules (primarily based on network traffic thresholds).	May produce false positives and may not take into account asset exposure or vulnerability context.
The proposed Hybrid Detection	makes use of ASM data, network traffic analysis, correlation logic and AI enhanced risk scoring.	Needs accurate ASM data and careful calibration of the risk scores.

3.11 Experimental Protocol

The experimental protocol will be the workflow as shown in Figure 3.2.

Figure 3.2 summarises the full workflow of the proposed methodology, from data collection and pre-processing through to feature extraction, ASM-traffic correlation, AI enhanced risk

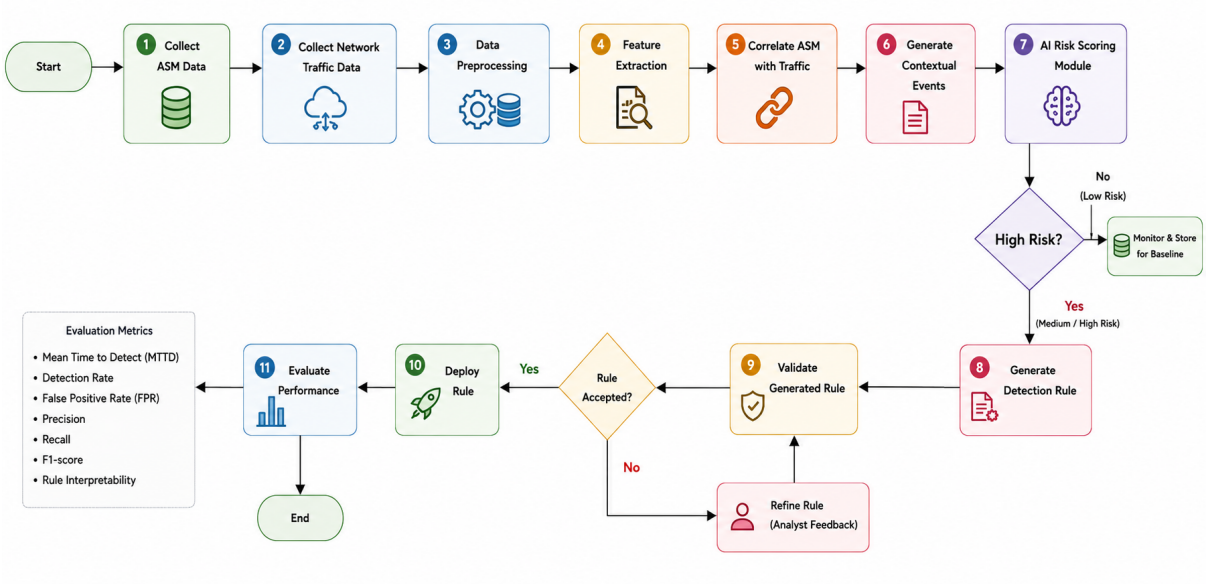


Figure 3.2: Proposed Methodology Flowchart

scoring, dynamic rule generation, rule validation, deployment and performance evaluation. The experimental procedure is as follows:

1. Prepare the ASM dataset (assets, services, ports, vulnerabilities, risk levels).
2. Prepare the network traffic dataset comprising the communication records and the various suspicious behaviour scenarios.
3. Apply some preprocessing to clean and normalise both datasets.
4. Extract relevant features from both the ASM and the traffic data.
5. Correlate the ASM information with the actual network traffic.
6. Apply AI enhanced risk scoring to prioritise the correlated events.
7. Generate dynamic detection rules for high risk scenarios.
8. Validate and deploy the generated rules.
9. Compare the proposed approach with the baseline detection approach.
10. Assess the methodology in terms of MTTD, Detection Rate, False Positive Rate, Precision, Recall, F1-score and Rule Interpretability.

This protocol provides a systematic and reproducible method for assessing the performance of the proposed hybrid approach against that of conventional rule-based detection methods.

3.12 Evaluation Metrics

The proposed method will be assessed using SOC relevant metrics (see Table 3.3). The use of a range of metrics is important as accuracy alone may not be sufficient to demonstrate the quality of detection, particularly where false positives and missed attacks are of operational significance [7]. Reducing detection time is important as incident response guidance specifically calls for improved efficiency and effectiveness in detection, response and recovery [11].

Table 3.3: Evaluation Metrics

Metric	**Purpose**
Mean Time to Detect	Measures the average time required to detect an attack or suspicious activity.
Detection Rate	Measures the percentage of actual attacks or suspicious scenarios detected by the system.
False Positive Rate	Measures the percentage of benign activities incorrectly classified as suspicious.
Precision	Measures how many generated alerts are actually relevant or correct.
Recall	Measures how many actual attack scenarios were successfully detected.
F1-score	Provides a balanced measure between precision and recall.
Rule Interpretability	Assesses whether the generated rule can be understood and reviewed by SOC analysts.

3.13 Experimental Environment

The experimental environment may include a laboratory network, public datasets, or simulated CSV-based datasets. The study may use tools such as Python for data processing, machine learning, risk scoring, and rule generation. The environment will document the following:

- Operating system.
- Programming language and libraries.
- Dataset sources.
- Preprocessing steps.
- Model or scoring settings.
- Evaluation scripts.

Documenting the experimental environment is key for reproducibility and for a fair evaluation.

3.14 Ethical Considerations

This study will of course avoid any unauthorised scanning, exploitation or collection of private network data. Any data used in the study will be taken from an authorised lab environment, a public dataset or simulated data created for the purposes of the research.

The study is not aimed at offensive exploitation. It is designed to improve defensive SOC detection by producing contextual and interpretable detection rules. The defensive focus of the study is in keeping with incident response and cyber risk management practice which is all about improving an organisation's preparedness, detection, response and recovery capability [11].

3.15 Threats to Validity

There are several threats to validity that could impact the study. The first is that lab or simulated data may not be representative of real SOC environments. Second, if the ASM data is inaccurate then the quality of the correlation and the rules that are generated will be compromised. Third, the AI risk scoring may need to be carefully calibrated to avoid any misleading results. Fourth, the results may be dependent on the attack scenarios that are used, the thresholds that are employed and the datasets that are used. These will be dealt with by clearly documenting the assumptions, using appropriate evaluation metrics, comparing against a baseline and clearly stating the limitations of the proposed methodology. Another validity threat is the accuracy of the asset and vulnerability data. If the ASM data is incomplete or out of date then the correlation process may lead to incorrect risk scores or detection rules. This threat will be addressed by fully documenting the data source, the data pre-processing and the assumptions that were made during the experiment [9].

3.16 Experimental Variables

This study has independent, dependent and control variables. It's important to define these variables in order to enable a clear and measurable assessment of the proposed methodology.

3.17 Proposed Timeline

The proposed timeline breaks the work down into phases. That timeline is of course subject to change depending on when the data becomes available, how the implementation progresses and what feedback we get from the supervisor.

Table 3.4: Experimental Variables

Variable Type	**Variable**	**Description**
Independent Variable	Detection method	(either traditional static detection rules or the proposed hybrid ASM based method).
Independent Variable	ASM context	(does the detection method use asset exposure, service risk, vulnerability severity and asset criticality?).
Dependent Variable	Mean Time to Detect	(the time it takes to detect the suspicious/malicious activity once it's in the traffic).
Dependent Variable	False Positive Rate	(the % of benign activity that is incorrectly classified as suspicious).
Dependent Variable	Precision, Recall and F1-score	(the metrics used to quantify the detection performance and quality of the alerts).
Control Variable	Dataset/Scenario	(the same dataset or attack scenario will be used when comparing the baseline to the proposed approach).
Control Variable	Time window	(the same time window will be used when comparing the results).
Control Variable	Threshold policy	(thresholds will be documented and applied in the same way).

Table 3.5: Proposed Research Timeline

Period	Planned Activity
Month 1	Finalise the research proposal, refine the research questions and confirm the scope with the supervisor.
Month 2	Carry out a detailed literature review on SOC detection, ASM, network traffic analysis, AI based detection and dynamic rule generation.
Month 3	Prepare the ASM data template, the network traffic data template and the structure of the experimental dataset.
Month 4	Implement the preprocessing steps and the ASM to traffic correlation logic.
Month 5	Develop the AI enhanced risk scoring and dynamic detection rule generation.
Month 6	Define the baseline detection rules and run the initial comparison experiments.
Month 7	Evaluate the proposed methodology using MTTD, false positive rate, precision, recall, F1 score and detection rate.
Month 8	Analyse the results, document the findings, discuss the limitations and refine the framework.
Month 9	Write the final thesis chapters and produce the tables, appendices and references.
Month 10	Final review, formatting, incorporate supervisor feedback, make corrections and prepare for submission.

3.18 Expected Outcomes

The expected outcomes of this study are:

1. A hybrid approach combining ASM data with network traffic analysis.
2. A correlation mechanism which correlates exposed assets with suspicious traffic behaviour.
3. An AI enhanced risk scoring component.
4. A dynamic detection rule generation process.
5. An assessment of detection time, false positives and alert accuracy.

3.19 Chapter Summary

This chapter details the research methodology and experimental design of the study. It covers the research design, the proposed framework, data sources and preprocessing, feature selection, the correlation logic, the AI-enhanced risk scoring, the dynamic rule generation, the baseline comparison, the experimental protocol, the evaluation metrics, the ethical considerations, the

threats to validity and the expected outcomes. The next section of the proposal is the conclusion and it sets the study up for implementation and evaluation. The study is also a defensive, privacy-aware piece of research in that it will not be carrying out any unauthorised exploitation, unauthorised scanning or collection of private traffic outside of the approved environment.

Conclusion

This research proposal describes a hybrid approach to the integration of real time Attack Surface Management (ASM) data with network traffic analysis in order to produce AI enhanced dynamic detection rules for the Security Operations Centre (SOC). The motivation for the work stems from the well documented shortcomings of current SOC detection methodologies which include delayed detection of attacks, high false positive rates, static detection rules and a poor correlation between asset exposure and actual network behaviour. The research problem identified is that there is no existing integrated and time aware methodology which brings together ASM data and network telemetry in order to enable the generation of contextual and interpretable detection rules. The proposed solution is a framework which utilises ASM data, network traffic features, correlation logic, AI enhanced risk scoring and dynamic detection rule generation. The framework is intended to be a support tool for the SOC analyst rather than a replacement for them. Its primary function is to improve the detection context by linking suspicious network traffic to the relevant exposed, vulnerable or critical assets. This will in turn assist the analyst in triaging alerts, reduce the number of false positives and enable earlier detection of attacks than is possible with the current generation of static rule based detection. The research will be conducted in accordance with a quantitative computing research design based on the evaluation of algorithms and systems. The proposed methodology will be benchmarked against a baseline detection approach which uses static or generic rules. The evaluation will be conducted using a range of SOC relevant metrics including Mean Time to Detect, false positive rate, precision, recall, F1-score, detection rate and rule interpretability. The overall expected contribution of the work is a well defined, practical and SOC focused methodology for the generation of dynamic detection rules from the combined evidence of ASM and network traffic. The research is expected to make a contribution to the field of more contextual, adaptive and explainable detection engineering in modern SOC environments.

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Appendix A

ASM Data Template

This appendix presents the proposed structure for Attack Surface Management (ASM) data used in the study. The purpose of this template is to standardize asset exposure information before correlating it with network traffic behavior.

Field	**Description**	**Example**
Asset ID	Unique identifier for the asset.	ASSET-001
Asset IP	IP address of the monitored asset.	192.168.1.10
Asset Name	Descriptive name of the asset.	Web Server
Service	Running service on the asset.	SSH, RDP, HTTP
Port	Open port associated with the service.	22, 3389, 80
Protocol	Transport protocol used by the service.	TCP or UDP
Exposure Status	Whether the service is exposed internally or externally.	Internal, External
Vulnerability Severity	Severity level of known vulnerability or weakness.	Low, Medium, High, Critical
Asset Criticality	Business or operational importance of the asset.	Low, Medium, High
Risk Level	Combined risk level based on exposure and criticality.	Medium, High, Critical

Appendix B

Network Traffic Data Template

This appendix presents the proposed structure for network traffic data used in the study. The purpose of this template is to standardize network behavior records before correlation with ASM data.

Field	Description	Example
Timestamp	Time when the traffic record was observed.	2026-01-21 10:30:00
Source IP	IP address initiating the connection.	185.10.10.5
Destination IP	IP address receiving the connection.	192.168.1.10
Destination Port	Port targeted by the connection.	22
Protocol	Network protocol used.	TCP
Connection Count	Number of connections within a time window.	50
Packet Count	Number of packets observed.	300
Byte Count	Number of bytes transferred.	25000
Time Window	Aggregation period for analysis.	60 seconds
Traffic Direction	Direction of communication.	External-to-Internal
Observed Behavior	Description of the traffic pattern.	Repeated SSH attempts

Appendix C

Dynamic Detection Rule Template

This appendix presents a generic template for detection rules generated by the proposed methodology. The generated rules should be interpretable and reviewable by SOC analysts before operational use.

Rule Element	Description
Rule ID	Unique identifier for the generated rule.
Rule Name	Short descriptive name of the rule.
Description	Explanation of the suspicious behavior detected by the rule.
Target Asset	The asset affected by the rule.
Target Service	The exposed or monitored service.
Detection Condition	Logical condition used to trigger the alert.
Threshold	Number or value required to trigger the rule.
Time Window	Time period used to evaluate the condition.
Risk Score	AI-enhanced or rule-assisted risk score assigned to the scenario.
Severity	Alert severity level such as Low, Medium, High, or Critical.
Recommended Action	Suggested action for SOC analysts.

Example Rule

Rule Name: Possible SSH Brute Force Against Exposed Asset

Condition: Alert when the same source IP attempts to connect to an exposed SSH service more than 20 times within 60 seconds.

Severity: High

Recommended Action: Review source IP reputation, validate authentication logs, check whether the destination asset is externally exposed, and block or investigate

the source if malicious behavior is confirmed.

Appendix D

Evaluation Checklist

This appendix provides a checklist for evaluating the proposed methodology during implementation and testing.

No.	Evaluation Item	Status / Notes
1	ASM dataset includes asset IP, service, port, exposure status, vulnerability severity, and asset criticality.	
2	Network traffic dataset includes source IP, destination IP, destination port, protocol, connection count, and time window.	
3	Data preprocessing steps are documented.	
4	ASM records are correctly mapped to traffic records.	
5	Correlation logic is clearly defined.	
6	AI-enhanced risk scoring method is explained.	
7	Generated detection rules are interpretable.	
8	Baseline detection approach is defined.	
9	Evaluation metrics are calculated.	
10	Threats to validity and limitations are documented.	

Appendix E

Risk Register Template

This appendix presents a risk register template for documenting possible risks during the implementation and evaluation of the proposed study.

Risk	Description	Impact	Mitigation
Incomplete ASM Data	Asset or exposure information may be missing or outdated.	High	Validate ASM data before correlation and document assumptions.
Noisy Traffic Data	Network logs may contain incomplete, duplicated, or irrelevant records.	Medium	Apply preprocessing and filtering before analysis.
False Positives	Generated rules may alert on benign behavior.	High	Use ASM context, thresholds, and analyst review to improve rule quality.
Poor Risk Score Calibration	AI or scoring model may overestimate or underestimate risk.	Medium	Test multiple thresholds and compare results with baseline rules.
Limited Dataset Realism	Lab or simulated data may not fully represent real SOC environments.	Medium	Clearly document dataset limitations and experimental assumptions.
Rule Interpretability Issue	Generated rules may be difficult for analysts to understand.	Medium	Use readable rule templates and include explanation fields.